

Linear Multistep Methods

$$u'(t) = f(u)$$

$$u(t_0) = \eta$$

An LMM takes the form:

$$\sum_{j=0}^r \alpha_j u^{n+j} = k \sum_{j=0}^r \beta_j f(u^{n+j})$$

We already know

$$u^n, u^{n+1}, \dots, u^{n+r-1}$$

we use this to find u^{n+r} .

If $\beta_r = 0$: explicit

If $\beta_r \neq 0$: implicit

Local truncation error

$$\sum_{j=0}^r \alpha_j u(t_n + jk) = k \sum_{j=0}^r \beta_j f(u(t_n + jk)) + k \tau^n$$

$$u(t_n + jk) = \sum_{i=0}^{\infty} \frac{(jk)^i}{i!} u^{(i)}(t_n) = u(t_n) + \sum_{i=1}^{\infty} \frac{(jk)^i}{i!} u^{(i)}(t_n)$$

$$f(u(t_n + jk)) = \sum_{i=0}^{\infty} \frac{(jk)^i}{i!} u^{(i+1)}(t_n) = \sum_{i=1}^{\infty} \frac{(jk)^{i-1}}{(i-1)!} u^{(i)}(t_n)$$

$$\sum_{j=0}^r \alpha_j u(t_n) + \sum_{j=0}^r \alpha_j \sum_{i=1}^{\infty} \frac{(jk)^i}{i!} u^{(i)}(t_n) - \sum_{j=0}^r \beta_j \sum_{i=1}^{\infty} \frac{j^{i-1} k^i}{(i-1)!} u^{(i)}(t_n) = k \tau^n$$

$$\frac{1}{k} \left[\sum_{j=0}^r \alpha_j u(t_n) + \sum_{j=0}^r \sum_{i=1}^{\infty} \frac{k^i j^{i-1}}{(i-1)!} u^{(i)}(t_n) \left(\alpha_j \frac{j}{i} - \beta_j \right) \right] = \tau^n$$

We want $\tau^n = \mathcal{O}(k^p)$
with p as large as possible.

$$\mathcal{O}(k^{-1}): \sum_{j=0}^r \alpha_j = 0$$

$$\mathcal{O}(k^0): \sum_{j=0}^r (j\alpha_j - \beta_j) = 0$$

Consistent

$$\mathcal{O}(k): \sum_{j=0}^r \left(j^2 \frac{\alpha_j}{2} - j \beta_j \right) = 0$$

2nd order

etc.

Examples: $\sum \alpha_j U^{n+j} = \sum \beta_j f(U^{n+j})$

2-step Adams-Bashforth

$$U^{n+2} = U^{n+1} + \frac{k}{2} (3f(U^{n+1}) - f(U^n))$$

$\alpha = [0, -1, 1]$

$\beta = [-\frac{1}{2}, \frac{3}{2}, 0]$

$\sum (j\alpha_j - \beta_j)$

$$= -1 + 2 - (-\frac{1}{2} + \frac{3}{2})$$

$$1 + \frac{1}{2} - \frac{3}{2} = 0$$

Leapfrog: $U^{n+2} = U^n + 2kf(U^{n+1})$

Backward differentiation
Formula (BDF2):

$$U^{n+2} = \frac{4}{3}U^{n+1} - \frac{1}{3}U^n + 2kf(U^{n+2})$$

For LMMs, consistency also
requires accuracy of the
starting values:

$$\lim_{k \rightarrow 0} \|U^j - u(t_j)\| = 0$$

for $j = 0, 1, \dots, r-1$.

Zero-Stability

If we apply a LMM to

$$U'(t) = 0$$

we get

$$\sum_{j=0}^r \alpha_j U^{n+j} = 0 \quad (*)$$

system of linear
difference eqns.

Ansatz: $U^n = \xi^n \quad \xi \in \mathbb{C}$

We have: $\sum_{j=0}^r \alpha_j \xi^j = 0$

We call $\rho(\xi) = \sum_j \alpha_j \xi^j$
the 1st characteristic
polynomial.

$\rho(\xi)$ is a polynomial of degree r
with roots $(\xi_1, \xi_2, \dots, \xi_r)$.

If these are distinct, then
all solutions of (*) take the
form

$$U^n = \sum_{j=1}^r c_n \xi_j^n. \quad \text{Notice that } \rho(1) = \sum_j \alpha_j = 0.$$

What about repeated roots?

For example: $U^{n+2} - 2U^{n+1} + U^n = 0$

$$y^{n+2} - 2y^{n+1} + y^n = 0$$

$$y^2 - 2y + 1 = 0$$

$$(y-1)^2 = 0$$

One solution: $U^n = 1^n = 1$

Another soln: $U^n = n y^n = n$.

$$\text{Check: } n+2 - 2(n+1) + n = 0$$

In general, a root ξ_j of multiplicity m leads to the fundamental solns:

$$U^n = \{ \xi^n, n \xi^n, n^2 \xi^n, \dots, n^{m-1} \xi^n \}$$

Again, the general solution is a linear combination of all fundamental solutions.

We want to know whether solutions of (*) remain bounded as $n \rightarrow \infty, k \rightarrow 0$, with $T = nk$ fixed.

We need:

- $|\xi_j| \leq 1$ for all j
 - If $|\xi_j| = 1$, it is a simple root.
- } The root condition

We say a LMM is zero-stable if $\rho(\xi)$ satisfies the root condition.

Theorem:

Any consistent zero-stable method is convergent for

$$\begin{aligned} u'(t) &= f(u) \\ u(t_0) &= \eta \end{aligned}$$

if f is Lipschitz continuous.

To prove this, we write the LMM as a one-step method:

$$V^n = \begin{bmatrix} U^n \\ U^{n+1} \\ \vdots \\ U^{n+r-1} \end{bmatrix}$$

$$V^{n+1} = \begin{bmatrix} U^{n+1} \\ U^{n+2} \\ \vdots \\ U^{n+r} \end{bmatrix}$$

The LMM

$$\sum_j \alpha_j U^{n+j} = K \sum_j \beta_j f(U^{n+j})$$

$$V^{n+1} = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & & \\ & & \ddots & \ddots & \\ & & & \ddots & 1 \\ \hline \frac{-\alpha_0}{\alpha_r} & \frac{-\alpha_1}{\alpha_r} & \dots & \frac{-\alpha_{r-1}}{\alpha_r} & \end{bmatrix} V^n + \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$

Companion matrix

$$\frac{K}{\alpha_r} \sum_j \beta_j f(U^{n+j})$$

The characteristic equation of C is $p(\xi)$.

$$V^{n+1} = CV^n + Kw^n$$

Substituting this into itself yields a formula with increasing powers of C .

$$\rightarrow E^{n+1} = CE^n + K\tau^n$$

So for convergence we need to bound

$$\|C^n\|_2 \text{ for all } n.$$

This is bounded iff $p(\xi)$ satisfies the root condition.